

BEAVERLODGE, AB, Canada

WMO: 712300

Lat: 55.1967N

Lon: 119.3964W

Elev: 745

StdP: 92.69

Time Zone: -7.00 (NAM)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.2	-27.5	-35.5	0.1	-30.8	-31.6	0.2	-27.3	13.9	3.5	12.3	3.3	1.3	40	0.689

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	26.9	15.0	25.0	14.3	23.3	13.7	16.6	23.6	15.6	22.4	14.7	21.1	3.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.2	11.0	18.6	12.9	10.2	17.4	11.9	9.5	16.8	49.4	23.7	46.1	22.8	43.5	21.1	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.3	9.6	8.3	-35.6	30.2	3.9	1.8	-38.4	31.6	-40.7	32.6	-42.9	33.7	-45.7	35.0		
			-35.8	18.3	3.8	1.4	-38.5	19.3	-40.8	20.1	-42.9	20.9	-45.7	21.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-10.6	-8.2	-4.7	3.2	9.9	13.6	15.8	14.7	9.9	3.5	-5.1	-9.7	(6)
(7)		DBStd	11.38	9.63	7.89	7.48	5.02	4.39	3.06	2.92	3.26	4.17	4.87	7.30	8.43	(7)
(8)		HDD10.0	3200	640	508	456	208	57	4	1	4	51	205	454	611	(8)
(9)		HDD18.3	5710	898	742	714	454	263	143	89	120	252	459	704	869	(9)
(10)		CDD10.0	552	0	0	0	3	52	114	180	148	49	5	0	0	(10)
(11)		CDD18.3	19	0	0	0	0	1	3	10	6	0	0	0	0	(11)
(12)		CDH23.3	376	0	0	0	1	21	53	158	114	28	1	0	0	(12)
(13)		CDH26.7	64	0	0	0	0	1	6	32	22	3	0	0	0	(13)
(14)	Wind	WSAvg	3.0	2.5	2.4	2.8	3.4	3.7	3.5	3.3	3.0	3.1	3.1	2.6	2.4	(14)
(15)	Precipitation	PrecAvg	500	38	25	25	22	42	77	68	68	50	26	31	32	(15)
(16)		PrecMax	799	126	78	51	57	159	183	175	160	130	58	79	129	(16)
(17)		PrecMin	324	6	2	3	3	2	14	12	14	3	3	0	0	(17)
(18)		PrecStd	108	29	18	14	17	33	50	42	39	30	14	22	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	7.1	10.8	20.0	25.8	27.2	30.0	29.2	26.7	20.0	11.9	8.0	(19)
(20)			MCWB	2.9	2.0	4.5	9.1	12.3	14.4	16.5	15.5	14.1	10.8	5.3	3.2	(20)
(21)		2%	DB	5.5	5.0	7.7	16.4	22.8	24.9	27.0	26.3	23.1	15.7	7.5	5.3	(21)
(22)			MCWB	1.8	1.1	2.6	7.4	11.1	13.6	15.4	15.4	13.1	8.4	3.6	1.8	(22)
(23)		5%	DB	3.9	3.5	5.9	13.4	20.6	22.7	24.9	24.1	20.2	12.8	5.2	3.4	(23)
(24)			MCWB	0.8	0.2	1.5	5.7	10.5	12.9	14.8	14.6	12.1	6.8	2.0	0.6	(24)
(25)		10%	DB	2.2	1.8	4.3	11.0	18.3	20.8	23.0	22.2	17.6	10.7	3.3	1.3	(25)
(26)			MCWB	-0.4	-1.1	0.5	4.3	9.3	12.1	14.1	13.8	10.7	5.6	0.7	-1.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.3	2.6	5.1	9.8	14.4	16.0	18.3	18.0	15.5	11.5	6.0	3.6	(27)
(28)			MCDWB	7.0	5.8	9.4	18.8	21.9	23.3	24.6	24.0	22.9	18.7	10.4	7.0	(28)
(29)		2%	WB	2.1	1.4	3.2	7.8	12.4	14.8	17.0	16.5	14.0	9.1	3.9	2.0	(29)
(30)			MCDWB	5.0	4.6	7.1	15.6	20.2	22.4	24.0	23.4	21.0	14.4	7.0	5.0	(30)
(31)		5%	WB	1.0	0.3	1.8	6.1	11.1	13.8	15.9	15.4	12.6	7.4	2.3	0.6	(31)
(32)			MCDWB	3.7	3.3	5.4	12.3	18.8	20.9	22.8	22.0	19.0	12.0	4.6	3.2	(32)
(33)		10%	WB	-0.4	-1.0	0.7	4.8	9.9	12.9	14.9	14.4	11.2	6.0	0.9	-1.0	(33)
(34)			MCDWB	2.0	1.8	4.1	10.4	17.2	19.1	21.0	20.7	17.1	10.1	3.2	1.2	(34)
(35)	Mean Daily Temperature Range	MDBR	8.2	8.6	9.1	10.2	11.8	11.3	11.6	12.0	11.3	9.1	7.4	7.8	(35)	
(36)		5% DB	MCDBR	8.9	9.0	10.2	14.7	15.6	14.5	15.2	15.8	16.1	13.0	9.5	8.6	(36)
(37)			MCWBR	6.7	6.3	6.6	8.0	7.3	6.2	6.4	6.7	7.7	7.4	6.7	6.8	(37)
(38)		5% WB	MCDBR	9.0	8.9	9.9	13.5	14.1	12.3	13.3	13.9	14.8	12.3	8.8	8.8	(38)
(39)			MCWBR	7.1	6.6	6.7	7.7	6.9	5.8	6.3	6.6	7.5	7.4	6.5	7.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.235	0.250	0.297	0.322	0.347	0.346	0.345	0.346	0.308	0.286	0.246	0.231	(40)	
(41)		taud	2.370	2.415	2.342	2.402	2.393	2.411	2.452	2.432	2.546	2.566	2.449	2.286	(41)	
(42)		Ebn at Noon	745	863	881	905	896	899	893	869	868	811	727	648	(42)	
(43)		Edh at Noon	48	67	93	102	109	108	102	98	76	59	45	41	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.75	1.67	3.11	4.62	5.47	5.98	5.87	4.79	3.31	1.79	0.82	0.54	(44)	
(45)		RadStd	0.07	0.13	0.24	0.28	0.39	0.48	0.39	0.41	0.31	0.19	0.07	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page